GG-S-161/2D
May 1, 1985
SUPERSEDING
GG-S-161/2C
May 3, 1977

FEDERAL SPECIFICATION SHEET

SCALES, DRAFTING: ENGINEER'S

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring Scales, Drafting: Engineer's, described herein shall consist of this document and the latest issue of GG-S-161.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers detail requirements for scales graduated in decimal parts of an inch to the inch, and scales graduated in the metric system, commonly called "chain scales".

1.2 Classification.

1.2.1 Type designation number. Scales covered by this specification shall be identified by a type designation number (see 6.2) formed as indicated in GG-S-161. The type designation number is intended for cataloging and ordering purposes and not for printing on the scales. Sub-paragraphs 1.2.1.3 through 1.2.1.10 indicate the options available for the type II scale and are further identified in GG-S-161.

1.2.1.1 Spec. No. GG-S-161.

1.2.1.2 Type of scales - -II.

1.2.1.3 Shape of scales - -1, -2, -5, -6, or -7.

1.2.1.4 Composition of scales - -A, -B, -C, or -D.

FSC 6675

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1.2.1.5 Grade of Scales - -1 or -2.

1.2.1.6 Size of scales - -B or -F.

1.2.1.7 Scale graduations - -1.

1.2.1.8 Numbering of scales - -D.

1.2.1.9 Style of scales - -A, -B, -C, -D, -E, -F, -G, -H, -J, -K, -L, -M, or -N (see 3.2).

1.2.1.10 Type of sheath required - -1, -2, -3, -4, -5, or -6.

3. REQUIREMENTS

3.1 Description. Scales shall be in accordance with GG-S-161 and as specified herein.

3.2 Graduations, shapes, and styles.

3.2.1 English system. Scales shall be fully divided and the divisions shall be boustrophedon numbered as follows: 10, every division marked to show inches only; 20, every 10th division numbered 1, 2, 3 ... etc.; 30, 40, 50, and 60, every 20th division numbered 2, 4, 6, ... etc.; 80 and 100, every 40th division numbered 4, 8, 12 ... etc.

3.2.2 Metric system. Scales shall be boustrophedon numbered as specified (see 1.2.1), fully divided, marked at each full centimeter, within the divisions and ratios as follows:

Scale divisions	Ratio
0.2000 mm	1:50
0.2500 mm	1:40
0.3333 mm	1:30
0.4000 mm	1:25
0.5000 mm	1:20
0.5999 mm	1:16.6667
0.6666 mm	1:15
1.0000 mm	1:10
2.0000 mm	1:5

3.2.3 Shapes and styles.

3.2.3.1 Shapes 1 and 2. Shapes 1 and 2 scales shall be as shown in figure 1 of GG-S-161, and shall be divided in the following pairs, one on each edge as specified by the following style letters (see 1.2.1.9).

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3.2.3.1.1 Style A. 10 and 50 parts to the inch.

3.2.3.1.2 Style B. 20 and 40 parts to the inch.

3.2.3.1.3 Style C. 30 and 60 parts to the inch.

3.2.3.1.4 Style D. 80 and 100 parts to the inch.

3.2.3.1.5 Style E. 5 and 25 parts to the centimeter.

3.2.3.1.6 Style F. 10 and 20 parts to the centimeter.

3.2.3.1.7 Style G. 15 and 30 parts to the centimeter.

3.2.3.1.8 Style H. 40 and 50 parts to the centimeter.

3.2.3.1.9 Style J. 16.6667 and 10 parts to the centimeter.

3.2.3.2 Shape 5. Shape 5 scales shall be as shown in figure 1 of GG-S-161, and shall be graduated as specified by the following style letters (see 1.2.1.9):

3.2.3.2.1 Style K. Style K scale shall be divided to tenths of an inch.

3.2.3.2.2 Style L. Style L scale shall be divided to fifths of a centimeter.

3.2.3.3 Shapes 6 and 7. Shapes 6 and 7 shall be as shown in figure 1 of GG-S-161, and shall be of the following styles (see 1.2.1.9).

3.2.3.3.1 Style M. Style M scales shall be graduated in the conventional divisions of 10, 20, 30, 40, 50, and 60 parts to the inch. The graduations shall be placed so that the scale is rolled over, the sequence shall be 10 - 20 - 30 - 10. The 50 shall be on the same side as the 10, the 40 as the 20, and the 60 as the 30.

3.2.3.3.2 Style N. Style N scales shall be graduated in divisions of 5, 10, 15, 20, 25, and 30 parts to the centimeter. The graduations shall be placed so that as the scale is rolled over, the sequence shall be 5 - 10 - 15 - 5. The 25 shall be on the same side as the 5, the 20 as the 10, and the 30 as the 15.

3.2.4 Scale length. Engineer's scales shall be furnished in 12-inch nominal lengths.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with GG-S-161.

5. PREPARATION FOR DELIVERY

5.1 Packaging shall be in accordance with GG-S-161.

6. NOTES. (The notes in GG-S-161 are applicable to this specification.)

6.1 Intended use. Engineer's scales are used in plotting and drawing maps.

6.2 Ordering data. Procurement documents should specify those desired options offered herein and should also specify the following:

a. Type designation number (see 1.2.1).

MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review activities

Army - GL
Navy - MC
DLA - GS

User activity

Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITIES:

FSS
NBS
OSS
SCS

PREPARING ACTIVITY:

Army - ME

Project 6675-0782

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